MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS

TYPE 5962

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Voltage regulator, corona discharge

See figure 1

Mounting position: Any

Weight: 0.3 ounce (8.5 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ez	Eb	Ib	TA	Barometric pressure, reduced
Unit:	Vdc	Vdc	µAdc	°C	mmHg
Maximum:	750	715	55	+75	---
Minimum:	---	685	5	-55	55
TEST CONDITIONS:	---	---	---	25 ±5	---

GENERAL:

Qualification - Required

(F) denotes changes

5962

FSC 5960

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3347	Ionization voltage	3		0.65	II	Ez	---	730	Vdc
3337	Voltage drop	-	Ebb/Ib = 25 ±2 µAdc	0.65	II	Eb(1)	690	710	Vdc
3335	Regulation	4	Ib = 5 to 55 µAdc	0.65	II	ΔEb	---	15	Vdc
---	Peak current	5	Ebb/Ib = 200 µAdc	0.65	II	---	---	---	---
---	Holding period end points:								
3347	Ionization voltage	6		0.65	II	Ez	---	730	Vdc
---	Change in voltage drop	6		0.65	II	ΔEb(1)	---	±3	Vdc
3335	Regulation	6		0.65	II	ΔEb	---	15	Vdc
---	Peak current	6		0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	1	No voltages applied	---	---	---	---	---	---
1026	Low-temperature operation:	-	T = -55°C						
3347	Ionization voltage	-		---	---	Ez	---	730	Vdc
---	Change in voltage drop	-		---	---	ΔEb(1)	---	+2, -5	Vdc
3335	Regulation	-		---	---	ΔEb	---	15	Vdc
---	Peak current	-		---	---	---	---	---	---
1026	High-temperature operation:	-	T = +85°C						
3347	Ionization voltage	-		---	---	Ez	---	730	Vdc
---	Change in voltage drop	-		---	---	ΔEb(1)	---	+2, +8	Vdc
3335	Regulation	-		---	---	ΔEb	---	15	Vdc
---	Peak current	-		---	---	---	---	---	---
3305	Leakage current	-	Eb = 500 Vdc (min); relative humidity = 90% +5%	---	---	Ilb	---	0.5	µAdc
	Quality conformance inspection, part 3								
---	Life test	-	Group B; Ebb/Ib = 55 ±5 µAdc; t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
3337	Voltage drop	-		---	---	Eb(1)	685	715	Vdc
3335	Regulation	-		---	---	ΔEb	---	15	Vdc
3347	Ionization voltage	-		---	---	Ez	---	730	Vdc
---	Peak current	-		---	---	---	---	---	---
1041	Shock	2,7	300 G	---	---	---	---	---	---
1002	Barometric pressure, reduced	7	Pressure = 55 mmHg; Eb = 750 Vdc	---	---	---	---	---	---

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NOTES:

1. Criterion for passing this test shall be compliance after test with the initial requirements for voltage drop and regulation.

2. Electrical samples shall not be used for this test. Ten additional samples shall be submitted for shock test only. Criterion for passing this test shall be compliance after test of at least 60 percent of the tubes with the initial requirements for voltage drop and regulation.

3. The tube under test shall not have been conducting for at least 30 minutes prior to test. During this test, the regulator tube shall not be exposed to external sources of radiation and shall be shielded from light. No conditioning current is permitted.

4. ΔEb maximum shall not be exceeded anywhere within the range 5 to 55 μAdc. In testing for regulation, current shall be varied continuously through the range of 5 to 55 μAdc.

5. Ebb shall be increased continuously until Ib = 200 μAdc. The voltage drop shall not go below its value at Ib = 55 μAdc.

6. The holding period commences upon completion of the initial performance of the specified tests. At the conclusion of the holding period, the specified tests will be performed again. Of the tubes failing the holding period end-point tests, those which fail with respect to voltage drop only, may be retested after an additional 30-day period. Eb(1) is the voltage drop measured at the beginning of a holding period.

(F) 7. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

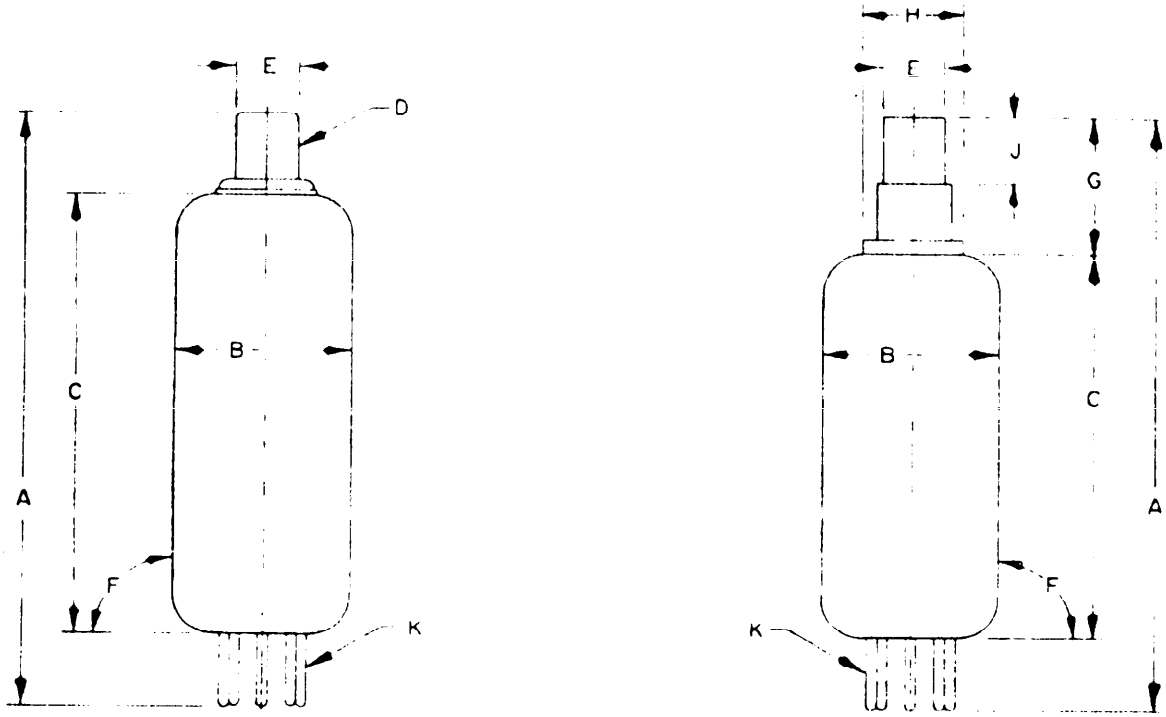
User activities:
Army - ME
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2993)

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ALTERNATIVE DRAWINGS

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	2.500 (63.50)	2.688 (68.28)
B		.750 (19.05)
C	1.703 (43.26)	2.094 (53.19)
Quality conformance inspection, part 3 (see note)		
D	Cap: Skirted miniature C1-3	
K	Base: E7-1	
E	.245 (6.22)	.255 (6.48)
F	86.5°	93.5°
Reference dimensions		
G	.600 (15.24)	
H	.420 (10.67)	
J	.281 (7.14)	

Ⓕ

PIN CONNECTIONS	
Pin No.	Element
1	k
2	k
3	k
4	k
5	k
6	k
7	k
cap	a

NOTE

Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 5962.

1975-- 603 766 1727

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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